Monochromator parts order

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1 Design objectives

Our monochromator system is meant to be a compact and reliable system, able to read wavelengths ranging from $600nm$ to $1.1\mu m$, with an adjustable resolution to gain in time-efficiency.

The physical design will have to fit on an 8”x10” or 10”x12” optical breadboard.

2 Diffraction grating

The main element of this design is the diffraction grating, which allows for a constant dispersion of light into its different wavelengths. The term constant is used to emphasize on the capacity of the monochromator to disperse light linearly, in contrast with some other dispersive optical elements such as the prism.

The diffraction grating is based on the principles of interference and diffraction:

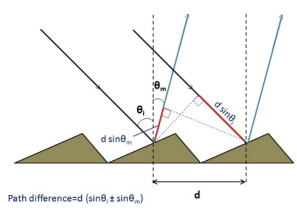

Figure 1: Path difference of two reflected lightwaves from the grating

For two given incoming lightwaves, the path difference between the two will be:

$$P = d * [\sin(\theta_i) \pm \cos(\theta_m)] \tag{1}$$

This motivates the driving monochromator equation, which defines the selected wavelength for a given angular configuration of the monochromator:

$$\sin(\alpha) + \sin(\beta) = 10^{-6} n \rho_{\text{groove}} \lambda \tag{2}$$

It is then possible to invert the equation to express α or β for a given target wavelength:

$$\alpha = \sin^{-1}\left[\frac{10^{-6} n \rho_{\text{groove}} \lambda}{2\cos(\frac{D_v}{2})}\right] - \frac{D_v}{2} \tag{3}$$

$$D_v \equiv \beta - \alpha \tag{4}$$

Here, α and β correspond to the angles between the incoming light rays and the normal of the monochromator:

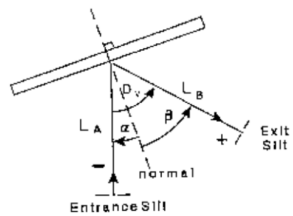

Figure 2: Representation of the dimensions in (3)

Notice that regardless of the angles α and β, D_v remains unchanged: it is one of the driving parameters of the design.

Finally, n and ρ_{groove} represent the diffraction order, and the grating's groove density, respectively. Since light intensity decreases with increasing order, we chose to only image at first order. The groove density has been defined based on the current availability of gratings on the market, as well as other important relations.

The angular dispersion for the monochromator is found to be:

$$\frac{d\beta}{d\lambda} = \frac{10^{-6} \rho_{\text{groove}} n}{\cos(\beta)} \tag{5}$$

Using Figure 2, we see that this corresponds to the following expression for linear dispersion:

$$\frac{d\lambda}{dx} = \frac{10^{6}\cos(\beta)}{\rho_{\text{groove}} L_B n} \tag{6}$$

With L_B being the arm length of the focused light.

Our goal is to have a system which satisfies $\frac{d\lambda}{dx}$ ranging from $1nm$ to $10nm$ for the aforementioned spectral range.

3 Design constraints

Since the diffraction grating used is reflection based, the incident and diffracted angles have to be between 0° and 90°. In terms of D_v, this means that the angle between the incoming and outgoing light must be between 0° and 180° (c.f. Figure 2). Therefore, the angle of deviation D_v, must be carefully chosen in order to satisfy this mechanical restriction.

By combining (3) and (6), one can obtain the linear dispersion as a function of the deviation angle for a given wavelength:

$$\begin{aligned} \frac{\partial \lambda}{\partial x}(\beta(D_v)) &= \frac{\lambda}{2L_B\gamma}\cos\left[\sin^{-1}\left(\frac{\gamma}{\cos(\frac{D_v}{2})}\right)+\frac{D_v}{2}\right] \\ \gamma &\equiv \frac{10^{-6}n\rho_{\text{groove}}\lambda}{2} \end{aligned} \tag{7}$$

It can be shown that the linear dispersion increases with increasing dispersion angle. However, the deviation angle is still limited by the capacity of the system to rotate. Indeed, most of the equations we have discussed are only defined for cases where the trigonometric functions, or their reciprocals, are defined. The angular range of a system is closely related to the diffraction grating groove density, the diffraction order, the central wavelength, as well as the deviation angle. Looking at equations (3) and (4), we see that the incident and diffraction angles are only defined for:

$$\begin{aligned} -1 \leq \frac{\gamma}{\cos(\frac{D_v}{2})} \leq 1 \\ \Leftrightarrow D_v \leq 2\cos^{-1}(\gamma) \text{ and } D_v \leq 2\cos^{-1}(\gamma) \end{aligned} \tag{8}$$

$\cos^{-1}(x)$ is a decreasing function. Therefore, the maximum deviation angle for our spectral range will be given by setting $\lambda = 1100nm$. In order to clearly define the maximum deviation angle, we must first discuss the diffraction order.

The goal is to limit the number of allowed diffraction orders to maximize intensity. Back to (2), it is possible to define the different allowable orders, given an incidence and deviation angle, by replacing the diffraction angle β by $D_v/2 + \alpha$.

$$D_v = \sin^{-1}[n10^{-6}\rho\lambda - \sin(\alpha)] - \alpha \tag{9}$$

Since $\sin^{-1}$ is only defined over -1, 1, we have an inequality for the term in the bracketed term involving the diffraction order. Say we would like the lower bound of these allowable order to be -1, and its upper bound to be 1, we find the range of desirable incident angles:

$$\begin{aligned} \frac{10^6}{\rho_{\text{groove}}\lambda}(\sin(\alpha)-1) \leq n \leq \frac{10^6}{\rho_{\text{groove}}\lambda}(\sin(\alpha)+1) \\ -1 \leq n \leq 1 \\ \alpha_1 = \sin^{-1}[-10^{-6}\rho_{\text{groove}}\lambda + 1], \text{ and } \alpha_2 = \sin^{-1}[10^{-6}\rho_{\text{groove}}\lambda - 1] \end{aligned} \tag{10}$$

This gives us $\alpha_{1,2}(600) = \pm 4.589°$ and $\alpha_{1,2}(1100) = \pm 78.52°$. It is now evident that we seek to only allow the -1st, 0th (uninteresting since it corresponds to simple reflection), and 1st orders of diffraction. Therefore, the maximum deviation angle is taken to be the smallest value from the inequalities in (8):

$$D_v^{\max} = 16.22° \tag{11}$$

Something else to consider is the influence on the aperture width. An exit slit represents a resolution cut off, which can allow for more coarse - and, most importantly, faster - measurements. We have previously mentioned the target resolution to range between $1nm$ and $10nm$. This means that, anywhere between $600nm$ and $1100nm$,

the slit should have the capability to be small enough for the $1nm$ resolution, but also wide enough to allow the $10nm$ resolution.

The expression for the slit width is taken from (6) and (7):

$$w = \frac{\Delta x}{\Delta \lambda}\Delta\lambda = \frac{2L_B\gamma}{\lambda}\left(\cos\left[\sin^{-1}\left(\frac{\gamma}{\cos(\frac{D_v}{2})}\right) + \frac{D_v}{2}\right]\right)^{-1}\Delta\lambda \tag{12}$$

For a qualitative understanding of this expression, we plot the curve $\frac{w}{\Delta\lambda}$ for $n = 1$ at both ends of the target spectrum.

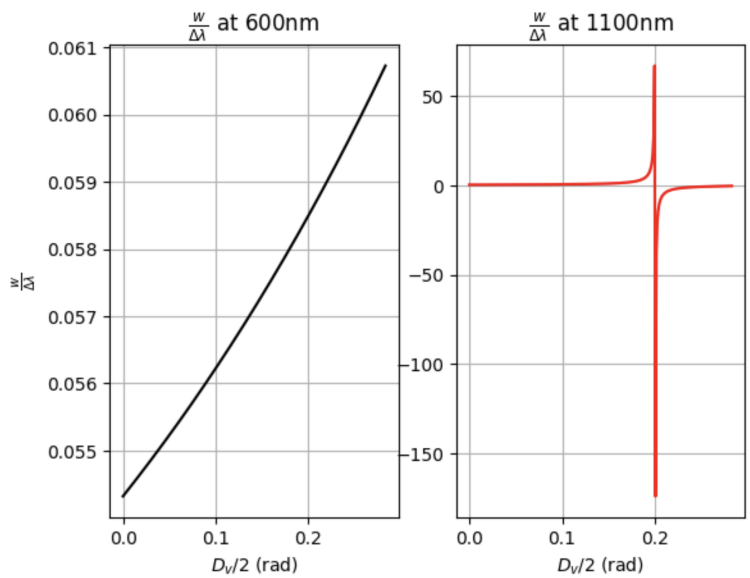

Figure 3: $\frac{w}{\Delta\lambda}$ vs $D_v/2$ at $\lambda = 600$, $\lambda = 1100$, and first order of diffraction

Observe that the relative slit width and the resolution increases with deviation angle, and that this increase itself leads to an increase with wavelength. However, at higher wavelengths, the ratio of the slit width and the resolution diverges. The deviation angle for divergence decreases until it reaches around $0.2rad$ at $1100nm$. Therefore, a deviation angle of $7.1^\circ = 0.12rad$ sounds reasonable in order to avoid any divergence of the dispersion, which renders spectral analysis impossible.

The last dimension to constrain is the arm length L_B, this dimension inversely scales with the linear dispersion, and can be adjusted in order to be able to implement a motorized slit. Still, we should seek to increase L_B as much as possible in order to increase the focal number of the parabolic mirror that it is related to.

4 Design outline

Now that we have better defined our system, we can go ahead and start arranging the various optical elements. For the purpose of calibration, four mirrors will be needed. Two for the monochromator, and two for the parabolic

mirror.

The deviation angle has been chosen to be $D_v = 7.1° < D_v^{\max}$. This deviation angle corresponds to a range of diffraction angles that can be found using (3) and (4): $\beta \in [36.30,\ 86.25]°$. Given this range of diffraction angles, the linear dispersion, in the first order, (6) gives the following range: $\frac{1}{L_B}[36.30,\ 447.71]nm/mm$. Therefore, for a $10nm$ resolution, (12) yields $w \in \frac{1}{L_B}[0.567,\ 7.00]mm$, while the slit width for $1nm$ ranges from $\frac{1}{L_B}0.0567mm$ to $\frac{1}{L_B}0.700mm$. Note that these are values for L_B in inches.

It is apparent that, with the smallest value for $w_{\min} = 0.0567mm = 56.7\mu m$, a small value of L_B seems relevant, which is why we have taken $L_B = 1in$, to finally define our entire system.

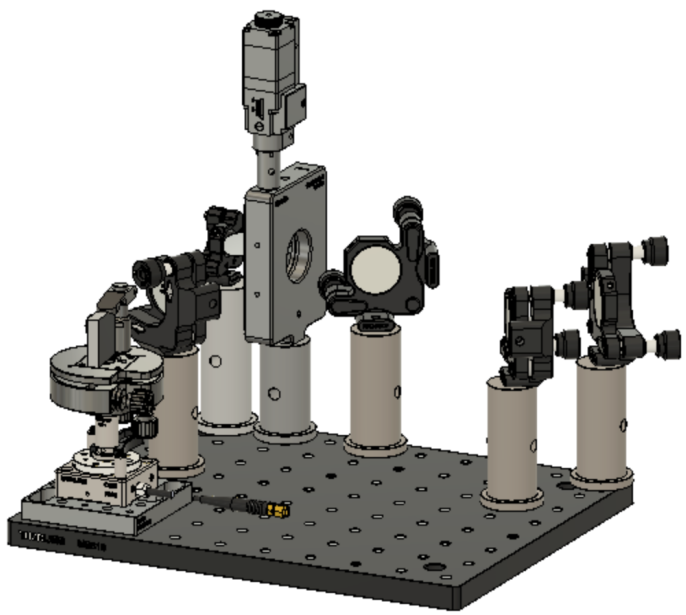

Figure 4: Monochromator system

Item	Qty	Price	Subtotal
Visible Reflective Holographic Grating 1800/mm 25 mm x 25 mm x 6 mm	1	159.18	159.18
Ø1” 90° Off-Axis Parabolic Mirror Prot. Aluminum RFL = 6”	1	194.83	194.83
Ø1” Ultrafast-Enhanced Silver Mirror 750 - 1000 nm	4	115.67	462.68
Kinematic Mirror Mount for Ø1” Optics, 8-32 Taps	4	119.97	479.88
ORIC Rotation Stage with Piezo Inertia Drive Imperial	1	944.09	944.09
Clamping Fork for 4-40 (M3) Cap Screw	1	17.82	17.82
Universal Mounting Adapter for ORIC Piezo Inertia Stages Imperial	1	227.46	227.46
Cage Assembly Rod 1” Long Ø6 mm	2	5.6	11.2
Motorized optical slit with linear actuator	1		0